

MOHAMMED SABOUR

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SUMMARY

Top-ranked graduate in Information Systems Engineering with self-motivation to pursue a Master of Science in Data Science. Experienced web developer with a strong desire to apply data science techniques to information technologies, such as natural language processing, big data systems, and machine learning. Extensive drive with ambition for data which was further amplified after achieving the highest mark in the 'Data Mining' module in a class of over 30 mathematics and computer science students. Solid foundation in Python programming, mathematics, and statistics, with hands-on experience in Big Data processing.

EDUCATION

Master's degree in mathematics and computer science Sep 2023 - Jun 2025

University of Mostaganem

- Specialization in information systems engineering.
- Graduated top of the class, 2025.
- Relevant modules : *Data analysis, Programming for Data Science, Advanced Programming Techniques, Data mining, Decision support systems.*
- Thesis on "Design and development of a collaborative Serious Game web platform".

Bachelor's degree in mathematics and computer science Sep 2020 - Jun 2023

University of Mostaganem

- Specialization in computer science.
 - Relevant modules : *Algebra, Numerical methods, Graph theory, mathematical logic, Algorithms and Data Structures.*
 - Thesis on "Development of a hospital management system (patients, staff, appointments)".
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WORK EXPERIENCE

Web Developer - Freelance work Oct 2024 -Dec 2024

Developed a web application to analyze Figma design files and automatically estimate development costs and timelines.

- Integrated the Figma API to extract interface design data.
- Automated data extraction via APIs and implemented intelligent component analysis and effort estimation using Gemini API.
- Built an interactive user interface to display dynamic estimation results.
- Implemented a configurable JSON system for customizable cost parameters.

IT Manager - Chamber of Trades and Crafts - Mostaganem Jul 2023 - Sep 2023

- Maintained and optimized IT hardware and software systems.
 - Installed and configured a local network (LAN) to improve internal connectivity.
 - Develop a barcode generator for inventory, optimizing stock tracking and management.
 - Installed card reader devices to streamline artisan registration.
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RELEVANT PROJECTS

Patient Clustering on Heart Disease Data

Dec 2024

Mini data mining project focused on unsupervised learning, applying K-Means clustering to medical (heart disease) data in order to identify patient groups based on similar health characteristics, without using labeled outcomes.

- Loaded and preprocessed a medical dataset related to heart disease.
- Cleaned the data by handling missing values and converting categorical data.
- Applied K-Means clustering to group patients based on similar clinical attributes.
- Determined the optimal number of clusters using the Elbow Method.
- Visualized and interpreted the results.

Big IOT data analysis with Hadoop

Dec 2024

Design of a Big Data system to analyze simulated environmental data using Python scripts, inspired by IoT sensors (temperature, humidity)

- Developed Python scripts simulating IoT sensors for massive data generation, ensuring efficient flow control and modular function organization
- Distributed storage with HDFS (Hadoop Distributed File System).
- Parallel processing with MapReduce (SensorMapper & SensorReducer) to identify correlations between metrics.
- Designed a master-slave distributed architecture with one master node and two slave nodes.

Worldwide COVID-19 Data Analysis & Visualization

Feb 2023

Mini data science project focused on time-series analysis of global COVID-19 data, processing pandemic datasets to track infection, recovery, and mortality trends across different regions.

- Loaded and preprocessed real-world COVID-19 datasets
- Performed data cleaning by handling missing values and ensuring consistent date-time formatting.
- Conducted time-series analysis of confirmed cases, recoveries, and deaths.
- Aggregated global and country-level metrics to study pandemic trends over time.
- Developed interactive visualizations using Plotly (time-series and scatter plots) to track the progression of COVID-19.

HARD SKILLS

- **Programming:** Python (PEP8, File handling, Debugging), SQL, C, Java.
- **Operating Systems:** Windows, Ubuntu (Linux).
- **Tools :** Anaconda, Jupyter Notebook, VSC, Git, Docker.
- **Databases & Web:** MySQL/PostgreSQL, MongoDB, Web APIs.
- **Data & AI :** NumPy, Matplotlib, Plotly, Pandas, Seaborn, Scikit-learn, Machine Learning.
- **Mathematics:** Calculus, Linear Algebra, Multivariate Analysis, Bayesian Statistics.

SOFT SKILLS

Communication | Teamwork | Problem-solving | Adaptability | Leadership | Creativity

ADDITIONAL INFORMATION

- **Languages:** English (B2) | French (B2) | Arabic (Native)
- **Interests :** Gaming, Travel, Football, Hiking, Fitness, Reading.